

Case Study: Brand Intelligence Agent

Time Budget: ~10–12 hours (we don't penalize going over — we do penalize unfocused effort)

Submission Window: 7 days from receipt **Final Step:** 30 minute live walkthrough with our team

Context

At Refabric, brand-level intelligence sits at the foundation of every product we ship. Before our AI can generate fashion designs that "feel like" a specific brand, we need a structured understanding of what that brand actually *is* — its visual signature, tone of voice, recurring silhouettes, color logic, audience cues, and content patterns.

Today, this work is largely manual. We want it to be automatic, repeatable, and scalable across thousands of brands. This case study is a focused slice of that problem.

Objective

Build a generic **Brand Intelligence Agent** that, given a fashion brand's identity (a name, a website URL, optionally social handles), autonomously produces a structured **Brand DNA dossier**.

The agent must be **brand-agnostic**: no hardcoded selectors per site, no per-domain scrapers. It must work on a brand it has never seen before.

Functional Requirements

1. Image Acquisition

- Collect **at least 100 images** from the brand's web presence that are confidently identified as **clothing or fashion-related**.
- Apply a quality filter: minimum resolution suitable for downstream AI processing (we suggest $\geq 512 \times 512$ on the shorter side — justify your choice).
- Visually deduplicate.

- Persist per-image metadata: source URL, surrounding text/alt, product name if extractable, page context, capture timestamp.

2. Multi-Source Data Collection

- **Brand website:** product pages, About, blog, lookbooks, press, FAQ.
- **At least one social channel** (Instagram is the priority; others optional).
- Be explicit in your documentation about how you handle anti-bot measures, rate limiting, and ToS considerations.

3. Brand DNA Synthesis (PDF Output)

Produce a Brand DNA report as a PDF that, at minimum, captures:

- **Visual identity:** dominant color palette, garment category mix, recurring silhouettes, styling cues (minimal vs. layered, casual vs. formal, etc.).
- **Textual identity:** brand voice, recurring vocabulary, stated values, positioning statement.
- **Audience signals:** who the brand appears to speak to (demographic, psychographic cues).
- **Aesthetic clustering:** 3–6 visual clusters with representative images and a short description per cluster.

The PDF must be readable by a non-technical brand strategist — **not a JSON dump dressed up as a document**.

4. Generalization & Repeatability

The same agent, with no code changes, must run successfully on at least **two brands of your choice** — ideally one mainstream e-commerce brand (well-structured site) and one smaller / less structured brand. This is the most important architectural test in the case.

Non-Functional Requirements

- Production-quality Python (or justify another stack).
- Tests covering critical logic (you decide what's critical — defend your choice).
- Containerized: Dockerfile, single command to run.
- Observability: structured logging, traceability of what the agent did and why.
- Graceful failure: site down, social blocked, no images found — none of these should crash the run.
- Configurable: a brand should be onboarded via config, not by editing code.

Deliverables

1. **Source repository** (GitHub link or archive).
 2. **Two complete sample runs** on two brands of your choice. Include the generated PDFs and the raw image/data outputs.
 3. **Architecture document** (1–3 pages): system diagram, key design decisions, trade-offs, what you'd build differently with more time and why.
 4. **README**: setup, how to run, configuration, dependencies, assumptions, known limitations.
 5. *Optional*: 5–10 minute walkthrough video.
-

Constraints & Latitude

- **Stack**: Python preferred. Library choice is yours.
 - **LLMs**: Free choice — Claude, GPT, Gemini, open-source. Justify your selection where it materially affects cost, latency, or quality.
 - **Hosted services** (Fal.ai, Replicate, OpenAI, etc.) are fine — explain trade-offs.
 - **Pretrained models** (CLIP, fashion classifiers, etc.) are encouraged. Don't train from scratch.
 - **Out of scope**: UI work, multi-tenancy, auth, deployment to production. A clean CLI is enough.
-

Evaluation Rubric

Dimension	Weight	What we look for
Architecture & generalization	25%	Site-agnostic design, clean separation of concerns, extensibility, configuration-driven
AI/ML approach	20%	Pragmatic model selection, sensible filtering/clustering logic, awareness of failure modes, cost-quality trade-offs

Dimension	Weight	What we look for
Code quality	20%	Readability, structure, testing, error handling, no dead code
Data engineering	15%	Storage layout, dedup strategy, metadata schema, throughput considerations
Documentation	10%	README clarity, architecture doc quality, design rationale
Output quality	10%	The PDF itself — does it actually convey brand DNA in a way a strategist can use?

What We're Explicitly NOT Looking For

- A polished UI. A CLI is fine.
- Exhaustive social coverage. One channel done well beats five done poorly.
- Heroic scraping. If a site blocks you, document it and move on.
- Over-engineering. We'd rather see a tight, well-reasoned scope than half-built ambition.
- Premature optimization. Make it work, make it clean, then talk about scale in the architecture doc.

What We Will Discuss in the Walkthrough

- Walk us through your architecture and the *why* behind each major decision.
- Live demo on a brand we pick (not one of your samples).
- Where would this break? What's the next iteration?
- How would you operationalize this for thousands of brands?

Submission

Send your repository link, the two sample outputs, and any supporting documents to safiye@refabric.com. We'll respond within 3 business days to schedule the walkthrough.

Good luck. We're looking forward to seeing your approach.